Pranav Joshi

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PROFESSIONAL SUMMARY

Data engineer and analyst with experience in large-scale SQL systems, Retrieval-Augmented Generation (RAG) pipelines, statistical modeling, Machine Learning (ML) and time series data. Skilled in end-to-end development from data ingestion to AI integration, with a strong foundation in computer science and applied mathematics.

TECHNICAL SKILLS

- **Languages:** Python, JavaScript, SQL, GAP, C, Verilog, MATLAB
- **Proficient in** Pandas, NumPy, SciPy, Matplotlib, Flask, Django
- **Worked with** opencv, socket, Selenium, sqlite3, oracledb, openai, ollama,
- **Databases:** Oracle, MySQL, SQLite
- **Frontend / UI:** HTML5 Canvas, Streamlit, Jinja, StencilJS
- **Developer Tools & Platforms:** Docker, Git, SSH, phpMyAdmin, PowerShell, Bash
- **Concepts:** RAG systems, AI agents, statistical modeling, time series data, data visualisation, Machine learning, Feature Engineering, Hypothesis testing

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EDUCATION

B.Tech in Computer Science and Engineering, IIT Gandhinagar
CGPA: 9.59 out of 10
Dean's List in semesters 2, 4, 5, and 6
Relevant Courses: Computer Vision, Machine Learning, Data Science, Databases, Signals & Systems, Biostatistics, Compilers, Computer Networks

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PROFESSIONAL EXPERIENCE

Data Intern, Texas Instruments — May 2025 to July 2025 Text2SQL — *June 2025 to July 2025*

- Conducted an in-depth literature review of Text-to-SQL and Large Language Models (LLM) based database querying approaches.
- Developed a metadata-free Retrieval-Augmented Generation (RAG) pipeline for column retrieval from enterprise SQL databases, handling **thousands of columns** and approximately **1 billion data points** without relying on metadata.
- Designed a system to estimate the "importance" of database columns for improved retrieval relevance.
- Built LLM-powered agents for table search, task decomposition, and SQL generation.
- Created a Streamlit application with StencilJS-inspired components and Redux-style state management via OOP in Python.
- Focused on scaling my Text2SQL application despite lack of metadata, which led to me developing a Jaccard Containment estimation technique suitable for large Oracle databases (see research section).
- **Technologies:** Python, Oracle SQL, SQLite, LlamaIndex, OpenAI API, Streamlit, StencilJS

End-of-Life Report Automation, — *May 2025 to June 2025*

- Automated the analysis of electronic parts flagged for end-of-life status, replacing manual review and reducing time spent from multiple days to minutes.
- Collaborated with business stakeholders to gather process requirements.
- Read source code for SpringBoot projects to understand the working of API endpoints.
- Reconstructed the same process using Oracle SQL queries.
- Implemented a GUI for the automation script and packaged it as an executable
- Solved many small issues of several employees related to SFTP and

Docker.

- **Technologies:** Python, Pandas, Oracle DB, Docker, StencilJS, Spring-Boot, SFTP

Awards and Achievements

- **PPO:** (Joining on July 13th) for the role of Data Engineer (not bond based).
- Presented in one of Tech-Fridays on topics such as Pandas and Numpy

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PROJECTS

SJL and Circadian Rhythm *(Computational Biology lab (CLOCK), IITGN, currently working)*

- Did exploratory data analysis on physiological fine grain and day-granularity data provided by Ultrahuman
- Applied data visualization techniques to identify two clusters in the data
- Used cosinor analysis with auto-regressive noise to capture the Circadian Rhythm
- Defined new metrics for social jet lag (SJL) such as heart-rate-autocorrelation (HRA)
- Characterised acute SJL events using changes in HRA

FPGA implementation of Total Focusing Method *(MUSE lab, IITGN, currently working)*

- Synthesised and Implemented a prototype on the Nexys 4 DDR board and later AC701
- Interfaced the FPGA with PC using UART and Ethernet (RMII/RGMII physical layer)
- Optimised by using dynamic programming, parallelism, pipelining, and streaming
- Created custom link layer protocol on top of Ethernet to reduce OS overhead.

New Website for Indian Drought Monitor *(WCL lab, IITGN, 25 May to 11 June 2026)*

- Created an interactive map showcasing CDI/SPI as a heatmap, using HTML5 canvas
- Created a local server for running LLM inference and a "Typst script to video" pipeline for automated summaries..

P4 based SRv6 and INT implementation and testing *(IITGN, Oct-Dec 2025)*

- Built a P4-based PDP, implementing IPv6 Segment Routing and SFANT style INT
- Wrote custom P4 parser and pipelines for BMv2 and integrated BMv2 with Mininet.
- Automated flow installation by running a shortest path algorithm on the controller.
- Benchmarked P4-based telemetry against SNMP/NetFlow.

Assamese News Corpus (Hugging Face Dataset) *(IITGN, Jan-Apr 2025)*

- Extracted and curated a large-scale Assamese language corpus from online newspapers (*NeoNow*, *Dainik Barta*, *Assam NeoNow*) using Python, Selenium, Hugging Face API.
- Published the dataset on Hugging Face for public use in low-resource NLP research.

Person Velocity Detection – Computer Vision Model *(IITGN, 2024)*

- Used YOLO-based object tracking and head detection models for person tracking.
- Used projective geometry and OpenCV to finally estimate the velocity

Open Source Contributions to GAP and SageMath *(IITGN, Jan-Jun 2024)*

- Collaborated with the author of a research paper on group theoretic algorithms.
- Reported bugs and implemented algorithms in GAP and SageMath.

Classical Machine Learning *(ES335, IITGN, 2023)*

- Implemented classic ML models like decision trees, with visualisations and ensemble models like random forests and used boot-straping, K-fold

split, etc., and read on optimisation methods like Momentum, SGD, batch gradient descent, Adam, etc.

Personal Projects

- Automatic differentiation system with backpropagation and neural network layers.
- Computational Physics Blog based on the course PH 509 at IITGN

Articles and Papers

Pranav Joshi, "Sampling-Based Estimation of Jaccard Containment and Similarity." *arXiv preprint, 2025.*

- Developed statistical models, Bayesian inference techniques, and bounds for set similarity estimation similar to the Lincoln-Peterson population size estimator.
- Empirically validated the models using simulated data.
- Gave guidance of applying this method for large databases with a rate/batch limit for querying.
- Applications: embedding propagation for RAG pipelines, plagiarism detection, literature search.